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Zhanhong He



Music Information Retrieval | Music Generation and Post-Training
University of Western Australia (UWA)

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Current Location: Hong Kong SAR, China

Summary

My research topic is Music and audio AI. Across my PhD, I developed models that connect music performance audio, symbolic data (MIDI, music scores) and musical expression. As visiting student researcher, I collaborate on the choral singing dataset creation, with the ChatGPT-based agentic workflow and Ace Studio synthesization. We also developed some MIR annotation works in this data curation progress; those were submitted as paper to ICASSP2027. During my Huawei internship, I worked on post-training music generation models (reinforcement learning, mainly on its Qwen3 component & flow-matching decoder). These works improved Huawei's internal models of its business project - YinYueJia [Webpage]. My doctoral defence at the University of Western Australia is completed, and the formal PhD award will arrive in December 2026.

Skills

Programming: Python / PyTorch / MATLAB (good); Arduino/ SQL / C / MaxMSP (fair)

PhD Research (MIR): Music Information Retrieval; Automatic Music Transcription; Psychoacoustic; Expressiveness Modelling; Music Source Separation; Differentiable Synthesizer (DDSP); Music Software.

Music Generation and Post-Training: VAE; Masked Transformer; Non-autoregressive generation; GRPO; REINFORCE++; music rewards; rollout workflows; human preference data (DPO); listening test design; training trajectory analysis.

Personal: Acoustic Guitar (intermedia); Collect music scores; Listen the Musical & Concert; HiFi; Driving; History; Game.

Education

PhD, Electrical and Electronic Engineering @ UWA **2022 Feb –2026 Dec**

- Supervised by **Prof. Roberto Togneri,** and co-supervised by **Prof. Defeng (David) Huang**.
- Thesis: ***Recovering Musical Dynamics for Expressive Automatic Music Transcription.***

BSc (Honours), Engineering Science @ UWA **2018 Feb –2021 Dec**

- First Class Honours; GPA 7/7; WAM 87.7/100 (top 3 in Electrical & Electronic department).
- Thesis: ***Improving Spectrogram-based Music Source Separation using Phase Information***.

Research and Industry Experience

Research Intern — Huawei Hong Kong Research Centre, Hong Kong SAR **Jun – Dec 2026**

- Applied DPO/GRPO to Qwen3 component for vocal/lyrics-to-music generation models. Analysed token-conditioned (Mucodec) and hidden-state-conditioned flow-matching decoding approaches and their implications for audio naturalness.
- Developed music rewards (e.g., PER-based lyric fidelity, beat alignment, and instrumental richness), and TRL & Miles rollout workflows. Utilised cross GPU notes' KL, data distribution and trajectory to diagnose & control post-training behaviour.
- Explored token-level RL policy optimisation (e.g., REINFORCE++), and applied Flow-GRPO in the decoding process. Improved final model internal MOS from 3.2/5 to 3.4/5, focused on audio quality & user prompt alignment.

Visiting Student Researcher — Great Bay University, Guangdong, China **Feb – Jun 2026**

- Hosted by **Prof. Yaolong Ju**. Developed LLM-agent dataset curation workflow, linking CPDL API to retrieve music score, and then using music21 to cleanup symbolic data, part-wise ACE Studio synth the dataset. [Link to PawChorale dataset].
- Developed a Semi-Markov decoder to assist beat/downbeat estimation. Submitted system best reported in MIREX2026.

Research Intern — Dolby Laboratories, Sydney, Australia **May – Sep 2024**

- Host by Staff Engineer David Cooper. Investigated U-Net, VAE and masked Transformer approaches to expressive music generation. Designed and conducted professional-listener experiments. Some works published in CMMR2025 paper.

Publications

Zhanhong He, Hanyu Meng, Jiahao Zhao, Ganghui Ru, Yaolong Ju, "MIREX 2026 Submission: Context-Aware Semi-Markov Post-Processor for Beat and Downbeat Estimation" in **ISMIR LBD**, 2026

Hanyu Meng, **Zhanhong He,** Yaolong Ju, "PawChorale: A Scalable Synthetic Dataset for Part-Aware Choral Transcription and Separation", in **ISMIR LBD**. 2026.

Zhanhong He, Hanyu Meng, David Huang, Roberto Togneri, "Beyond Piano: Cross-Instrument MIDI Velocity Estimation via Differentiable SoundFont Proxies" in Int. Soc. Music Info. Retrieval Conference (**ISMIR**), 2026.

Zhanhong He, Roberto Togneri, David Huang, "Score-informed Transformer for Refining MIDI Velocity in Automatic Music Transcription" in Sound and Music Computing (**SMC**), 2026.

Zhanhong He, Hanyu Meng, David Huang, Roberto Togneri, "Joint Estimation of Piano Dynamics and Metrical Structure with a Multi-task Multi-Scale Network" in IEEE Int. Conference Acoust., Speech, Signal Process. (**ICASSP**), 2026.

Zhanhong He, David Cooper, David Huang, Roberto Togneri, "Filling the MIDI Velocity using U-Net Image Colorizer" in Int. Symp. on Computer Music Multidisciplinary Research (**CMMR**), 2025.

(Submitted) Zhanhong He, Hanyu Meng, Yaolong Ju, "CASM: Context-Aware Semi-Markov Post-Processor for Beat Tracking" under review in ICASSP2027.

(Submitted) Hanyu Meng, **Zhanhong He,** Zixun Guo, Yaolong Ju, "Toward Part-Aware Choral Transcription with singing voice assignment" under review in ICASSP2027.

(Submitted) Hanyu Meng, Hao Shi, **Zhanhong He**, … "Speaker-Consistent Grounding for Confusion-Resilient LLM-Based Target Speech Extraction" under review in ICASSP2027.

(Submitted) Mingda Song, Yueqiao Zhang, Yulun Wu, **Zhanhong He,** … "HCMAMBA: Harmonically Informed State-Space Modeling for Large-Vocabulary Audio Chord Recognition" under review in ICASSP2027.

Selected Research and Music Course Project

Beyond Piano: cross-instrument expression control [Code]

- Developed a differentiable SoundFont-proxy approach for adapting piano-pretrained MIDI velocity estimator to other instruments, including guitar, using loudness-related audio supervision.

Estimate piano dynamics and metrical structure (beat, downbeat) [Code]

- Implemented Bark-scale loudness features in PyTorch using established psychoacoustic methods; worked on multitask multi-scale models (MSCNN) of dynamic levels, change points, beats and downbeats.

CASM Beat tracking post-processor [Code]

- Developed a context-aware semi-Markov (CASM) decoder for neural beat estimators; verified its effectiveness on BeatThis (Transformer), MSCNN, and TCN; achieved best performance in MIREX2026 Beat/Downbeat challenges.

Electronic Projects @ UWA (Coursework Designs) [Technical Reports]

- **Simple Theremin:** Built on breadboard w. Arduino, changes pitch when moving hand (i.e., light-controlled resistant). **Bicycle Odometer:** Simulated on TinkerCAD, requested PCB manufacturing, soldered Arduino Nano chip on PCB, programming it with C Language, and finally 3D-printed its enclosure. **Workstation Monitor:** Similar to above; instead of using a Arduino on board, we migrated its code into an ATmega328P microcontroller.

Conservatorium of Music @ UWA (Coursework Designs)

- **MUSC2981 Music Ensemble:** Form a 13 members team to perform Marimba in a Public Concert [YouTube Video]
- **MUSC2273 Music Generation:** Design an HMM-based Auto Piano and Environment Sounds generative system, programming in Max/MSP with Ableton Live. [Demo Video]
- **MUSC1054 Electronic Music:** Use Ableton Live to make music composition & mixing.

Teaching and Awards

UWA Tutor/Lab Facilitator, 2023 − 2025: Signal Processing, Electronic Laboratories; co-supervised CS honours projects.

Awards: University International Fee Scholarship; University Postgraduate Award; 2026 UWA Data Institute Award; 2018 International Exchange Scholarship; 2017 Excellent Student Club President.

Referees: Prof. Roberto Togneri (UWA) roberto.togneri@uwa.edu.au; Prof. Yaolong Ju (GBU, China) juyaolong@gbu.edu.cn